

Clinical Risk Assessment Report: Patient 40

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 57.5% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 2.60x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +7.0%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 0.87 [Top 11%] (+7.0%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.37 [Bottom 9%] (+6.6%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.83 [Top 20%] (+4.8%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: 0.95 [Top 16%] (+4.2%)**
Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 0.9 [Top 19%] (+4.2%)**
Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.71 [Top 23%] (+3.6%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.25 [Bottom 5%] (+2.9%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: 1.02 [Top 15%] (+2.8%)**
Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

